

LIGENTEC Generic Layout Manual

Version 8.9

Generic Layout Manual - CONTENTS

1. Generic layout definitions.....	3
2. Optical I/O structures.....	4
2.1. ExSpot for SMF fibre black box.....	4
2.2. ExSpot for packaging SMF fibre black box.....	5
2.3. Edge Coupler for Lensed fibre black box.....	6
2.4. Focused Grating Coupler black box.....	7
3. Power Splitter structures.....	8
3.1. 1x2 MMI symmetric black box.....	8
3.2. 2x2 MMI symmetric black box.....	9
3.3. 2x2 MMI 85-15 black box.....	10
3.4. 2x2 MMI 72-28 black box.....	11
3.5. 2x2 MMI 90-10 black box.....	12
3.6. 2x2 MMI 95-05 black box.....	13
3.7. 3x3 MMI black box.....	14
3.8. 4x4 MMI black box.....	15
4. Crossing waveguides.....	16
4.1. Crossing Ultra Compact TE black box.....	16
4.2. Crossing TE black box.....	17
4.3. Crossing Balanced black box.....	18
5. Polarization Management.....	19
5.1. Polarization Beam Splitter black box.....	19
6. Others.....	20
6.1. Optical Via Black Box.....	20
7. Change Log.....	21

1. Generic layout definitions

This section contains all the necessary information to draw all the proprietary Ligentec building blocks so any software can be used for layout. For every building block you will find a table including information about the correct name to use, the optical and electrical port positions, the port angle, the box boundary positions, and the layer to use for each port.

For the latter, all the building block boxes have been aligned following the next nomenclature, with **in0** always located at **(x, y) = (0, 0)**:



where BIB and BOT represent **B**ox **I**nput **B**ottom and **B**ox **O**utput **T**op respectively.

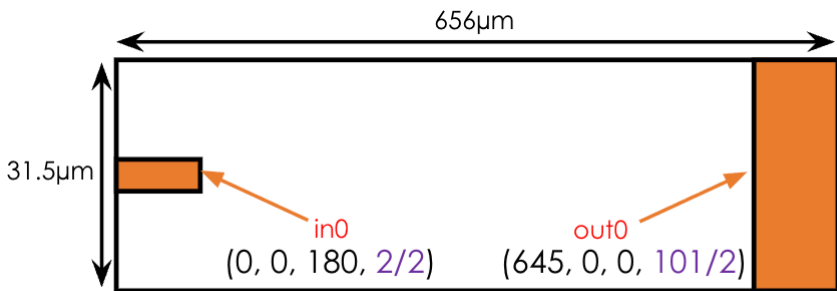
Leaving building block names unchanged is very important since the replacement of the black boxes before fabrication will depend on that name. Also, it is necessary to remark that **hierarchy should be preserved in the layout. If not, the building block replacement will fail.**

In the following sections, the position, angle, and layer of the ports are indicated in the following format: (X-coordinate in μm , Y-coordinate in μm , outward facing angle in degrees, port layer).

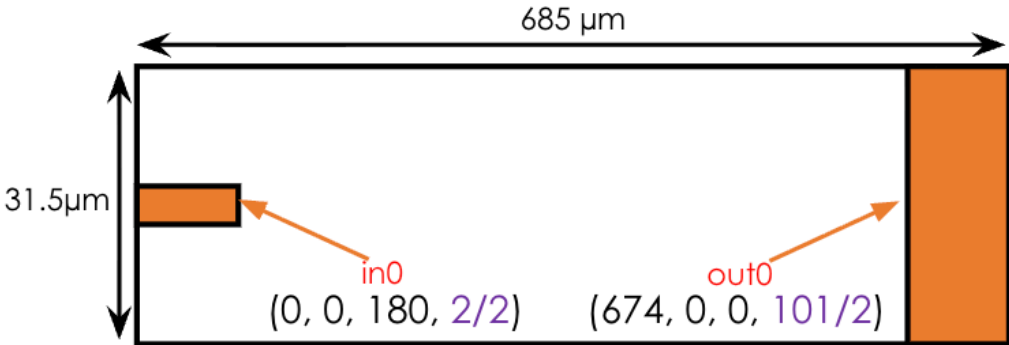
Please, refer to the Design Manual Section 3.3.1 for connecting a waveguide to a black box.

2. Optical I/O structures

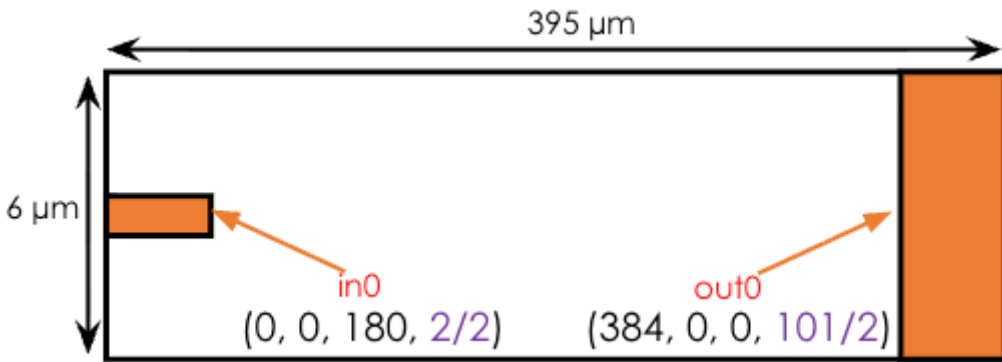
2.1. ExSpot for SMF fibre black box

Description	
Black box name	AN800BB_ExSpot_SMF_C
Maturity level	Beta
Required modules	X1, ExSpot
Dimensions	
Box size Input waveguide widths	656 μm x 31.5 μm 1 μm
Ports	
Optical ports Electrical ports	in0, out0 none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_ExSpot_SMF_C	
BIB and BOT	
BIB BOT	[-1.0, -15.75, 0] [655.0, 15.75, 0]

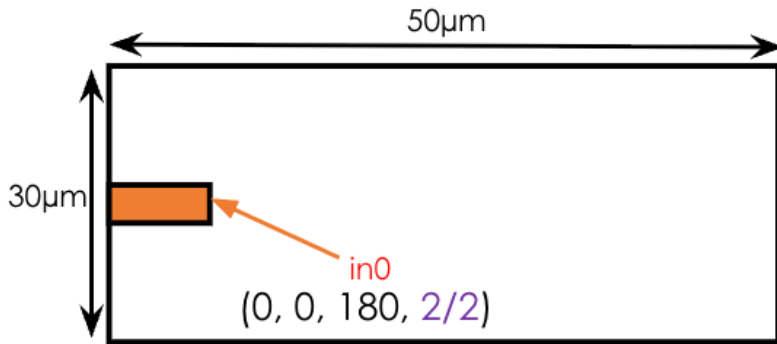
2.2. ExSpot for packaging SMF fibre black box

Description	
Black box name	AN800BB_ExSpot_packaging_SMF_C
Maturity level	Beta
Required modules	X1, ExSpot
Dimensions	
Box size	685 μm x 31.5 μm
Input waveguide widths	1 μm
Ports	
Optical ports	in0, out0
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_ExSpot_packaging_SMF_C	
BIB and BOT	
BIB	[-1.0, -15.75, 0]
BOT	[684, 15.75, 0]

2.3. Edge Coupler for Lensed fibre black box

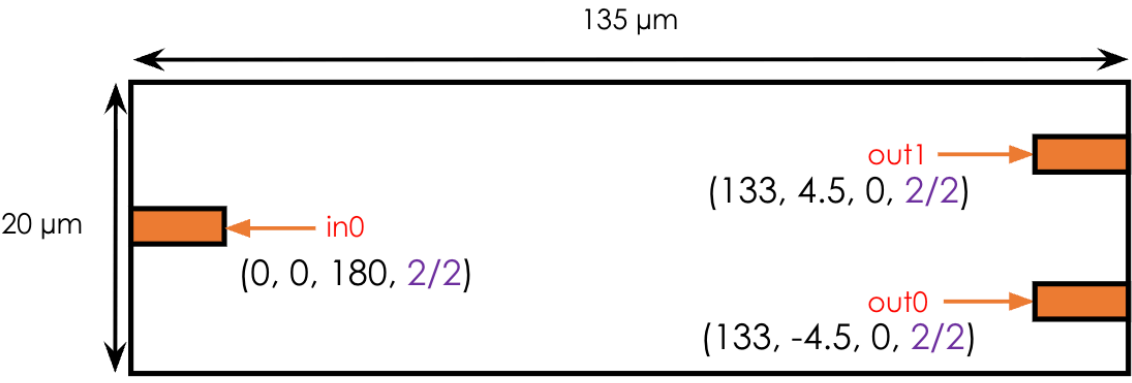
Description	
Black box name	AN800BB_EdgeCoupler_Lensed_C
Maturity level	Released
Required modules	X1
Dimensions	
Box size	395 μm x 6 μm
Input waveguide widths	Variable width
Ports	
Optical ports	in0, out0
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_EdgeCoupler_Lensed_C_wX.Y , where X.Y will be the input waveguide width in μm .	
BIB and BOT	
BIB	[-1.0, -3.0, 0]
BOT	[394.0, 3.0, 0]

2.4. Focused Grating Coupler black box

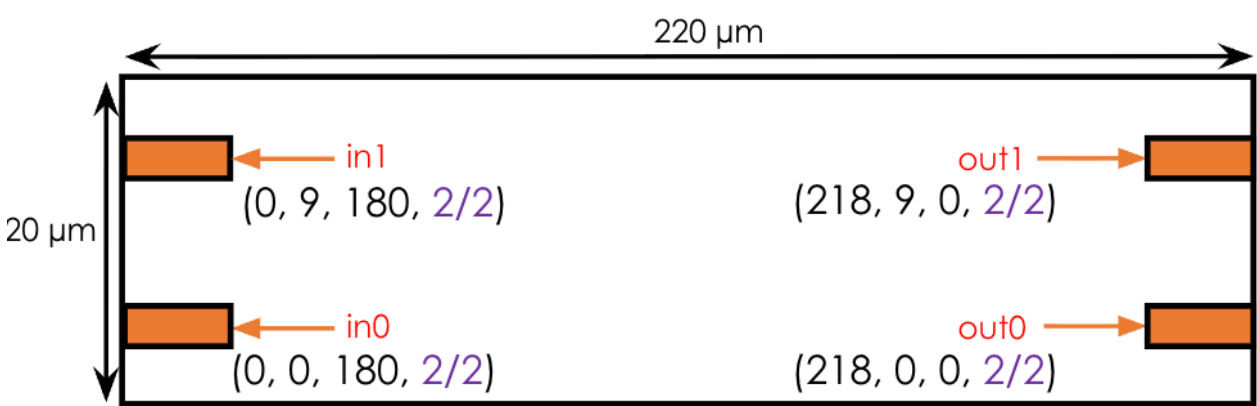
Description	
Black box name	AN800BB_FGC_8_TE_C
Maturity level	Beta
Required modules	X1
Dimensions	
Box size	50 μm x 30 μm
Input waveguide widths	1 μm
Ports	
Optical ports	in0
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_FGC_8_TE_C	
BIB and BOT	
BIB	[-1.0, -15.0, 0]
BOT	[49.0, 15.0, 0]

3. Power Splitter structures

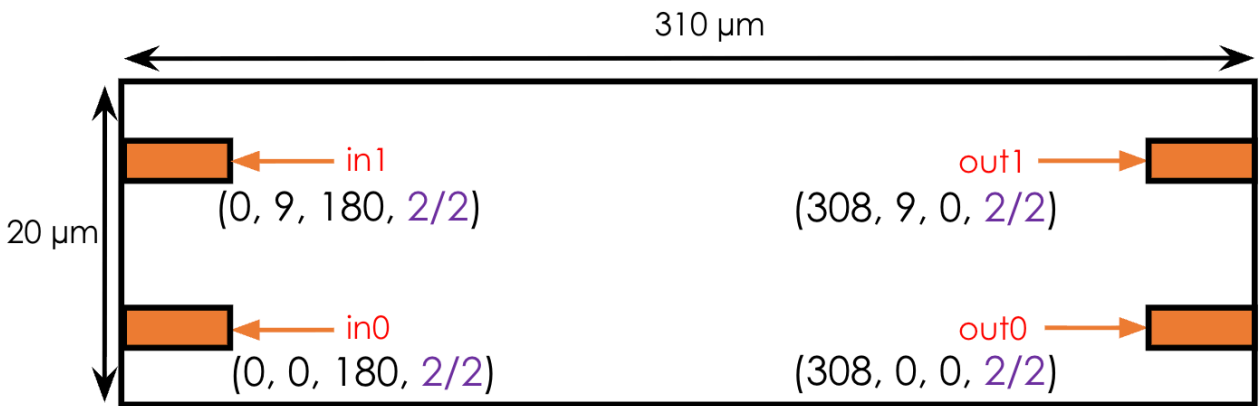
3.1. 1x2 MMI symmetric black box

Description	
Black box name	AN800BB_MMI1x2_symmetric_C
Maturity level	Released
Required modules	X1
Dimensions	
Box size	135 μm x 20 μm
Input and output waveguide widths	1 μm
Ports	
Optical ports	in0, out0, out1
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_MMI1x2_symmetric_C	
BIB and BOT	
BIB	[-1.0, -10.0, 0]
BOT	[134.0, 10.0, 0]

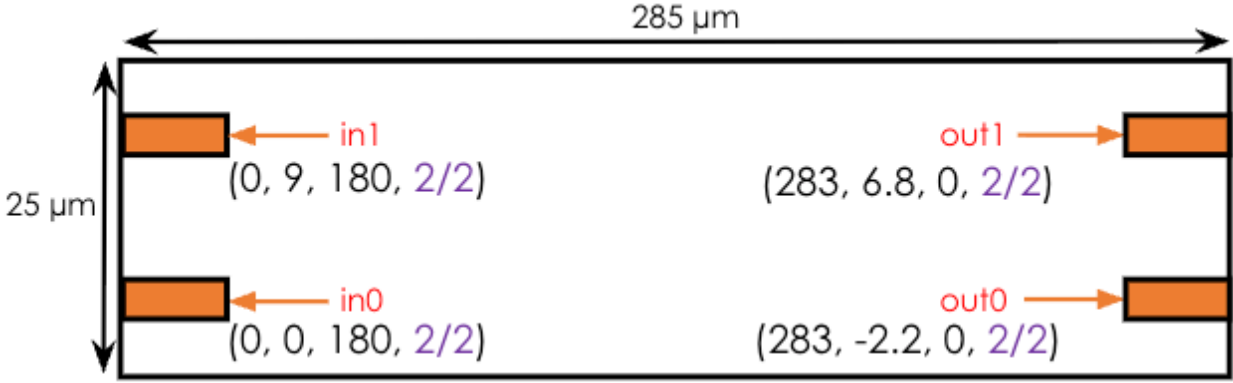
3.2. 2x2 MMI symmetric black box

Description	
Black box name	AN800BB_MMI2x2_symmetric_C
Maturity level	Released
Required modules	X1
Dimensions	
Box size	220 μm x 20 μm
Input and output waveguide widths	1 μm
Ports	
Optical ports	in0, in1, out0, out1
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_MMI2x2_symmetric_C	
BIB and BOT	
BIB	[-1.0, -5.5, 0]
BOT	[219.0, 14.5, 0]

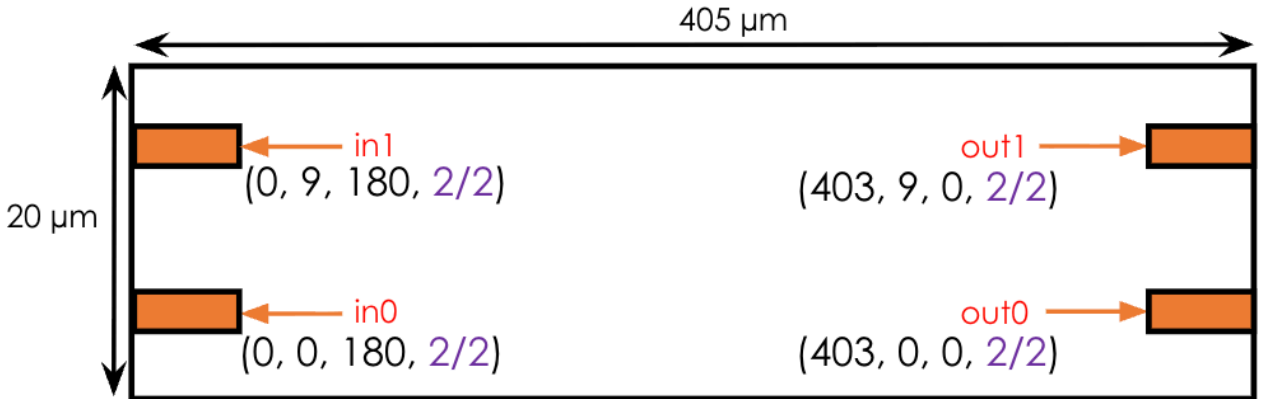
3.3. 2x2 MMI 85-15 black box

Description	
Black box name	AN800BB_MMI2x2_85_15_C
Maturity level	Released
Required modules	X1
Dimensions	
Box size	310 μm x 20 μm
Input and output waveguide widths	1 μm
Ports	
Optical ports	in0, in1, out0, out1
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_MMI2x2_85_15_C	
BIB and BOT	
BIB	[-1.0, -5.5, 0]
BOT	[309.0, 14.5, 0]

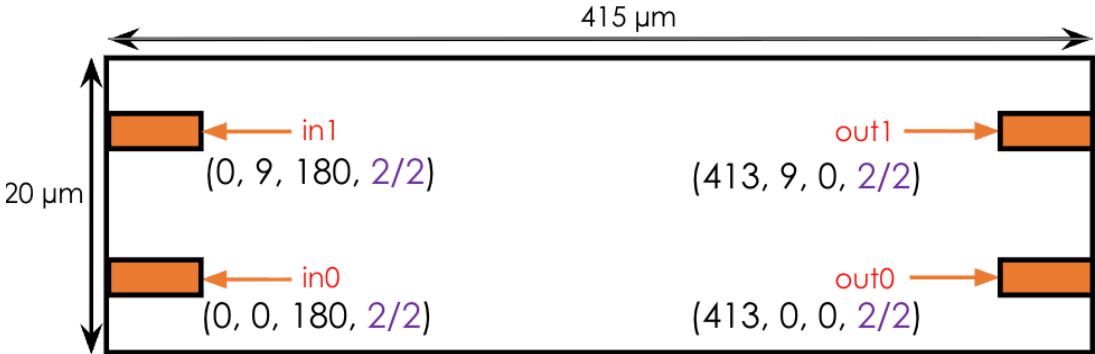
3.4. 2x2 MMI 72-28 black box

Description	
Black box name	AN800BB_MMI2x2_72_28_C
Maturity level	Released
Required modules	X1
Dimensions	
Box size	285 μm x 25 μm
Input and output waveguide widths	1 μm
Ports	
Optical ports	in0, in1, out0, out1
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_MMI2x2_72_28_C	
BIB and BOT	
BIB	[-1.0, -7.7, 0]
BOT	[284.0, 17.3, 0]

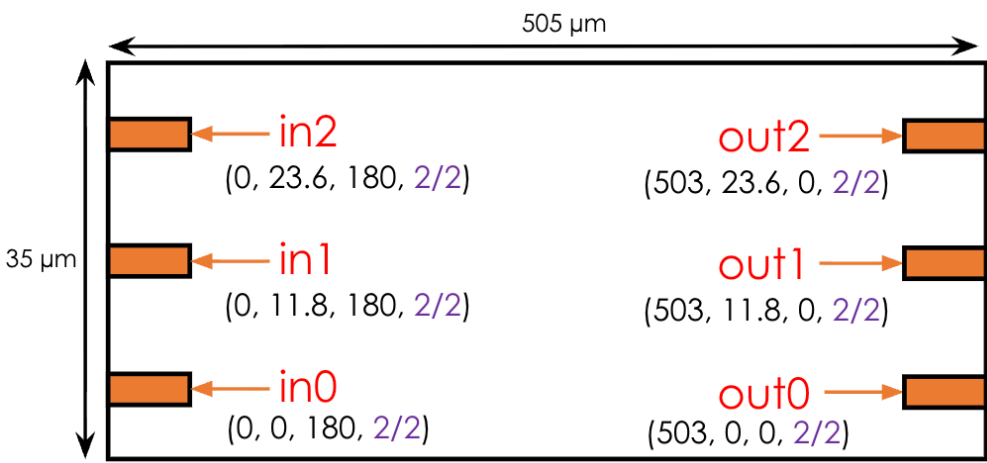
3.5. 2x2 MMI 90-10 black box

Description	
Black box name	AN800BB_MMI2x2_90_10_C
Maturity level	Beta
Required modules	X1
Dimensions	
Box size	405 μm x 20 μm
Input and output waveguide widths	1 μm
Ports	
Optical ports	in0, in1, out0, out1
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_MMI2x2_90_10_C	
BIB and BOT	
BIB	[-1.0, -5.5, 0]
BOT	[404.0, 14.5, 0]

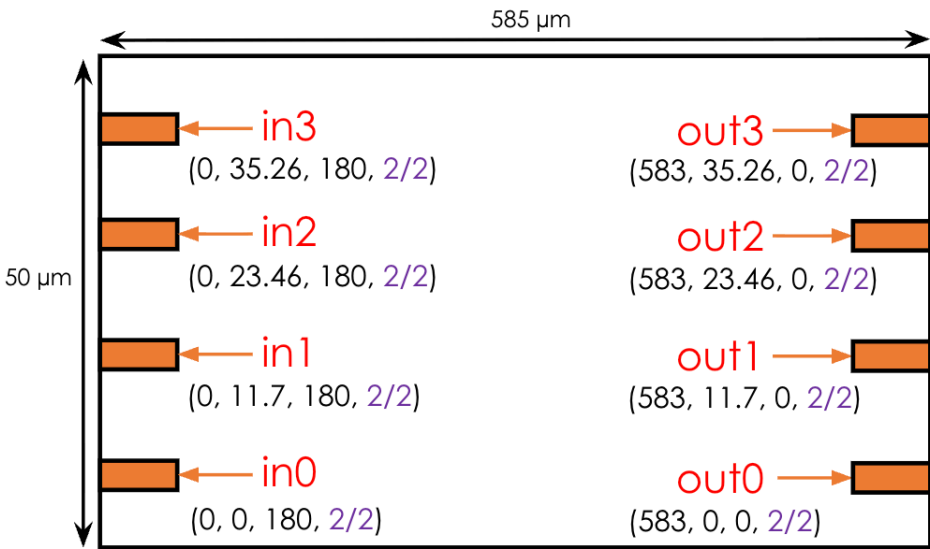
3.6. 2x2 MMI 95-05 black box

Description	
Black box name	AN800BB_MMI2x2_95_05_C
Maturity level	Alpha
Required modules	X1
Dimensions	
Box size	415 μm x 20 μm
Input and output waveguide widths	1 μm
Ports	
Optical ports	in0, in1, out0, out1
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_MMI2x2_95_05_C	
BIB and BOT	
BIB	[-1.0, -5.5, 0]
BOT	[414.0, 14.5, 0]

3.7. 3x3 MMI black box

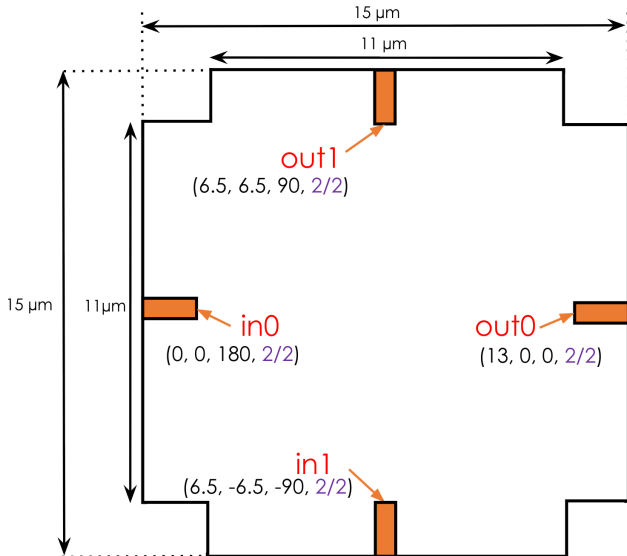
Description	
Black box name	AN800BB_MMI3x3_symmetric_C
Maturity level	Alpha
Required modules	X1
Dimensions	
Box size	505 μm x 35 μm
Input and output waveguide widths	1 μm
Ports	
Optical ports	in0, in1, in2, out0, out1, out2
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_MMI3x3_symmetric_C	
BIB and BOT	
BIB	[-1.0, -5.7, 0]
BOT	[504.0, 29.3, 0]

3.8. 4x4 MMI black box

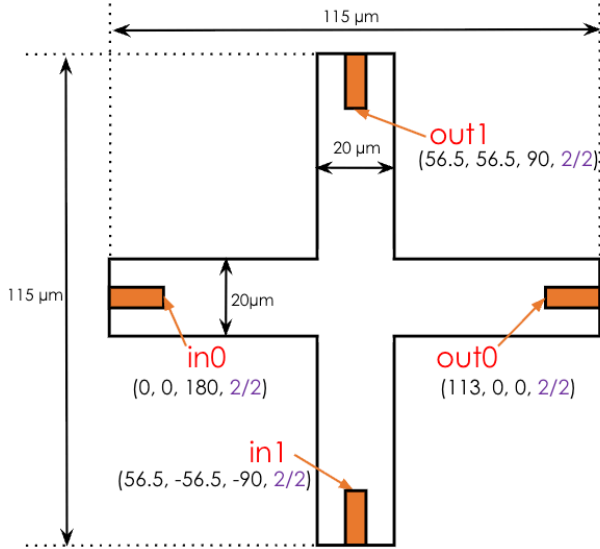
Description	
Black box name	AN800BB_MMI4x4_symmetric_C
Maturity level	Alpha
Required modules	X1
Dimensions	
Box size	585 μm x 50 μm
Input and output waveguide widths	1 μm
Ports	
Optical ports	in0, in1, in2, in3, out0, out1, out2, out3
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_MMI4x4_symmetric_C	
BIB and BOT	
BIB	[-1.0, -7.42, 0]
BOT	[584.0, 42.58, 0]

4. Crossing waveguides

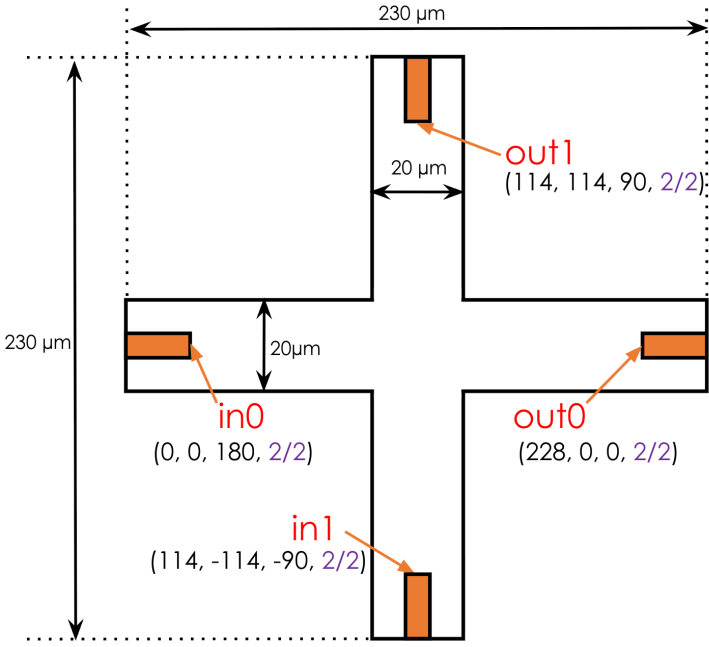
4.1. Crossing Ultra Compact TE black box

Description	
Black box name	AN800BB_Crossing_UltraCompactTE_C
Maturity level	Released
Required modules	X1
Dimensions	
Box size	Refer to schematic
Input and output waveguide widths	1 μm
Ports	
Optical ports	in0, in1, out0, out1
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_Crossing_UltraCompactTE_C	
BIB and BOT	
BIB	Refer to schematic
BOT	Refer to schematic

4.2. Crossing TE black box

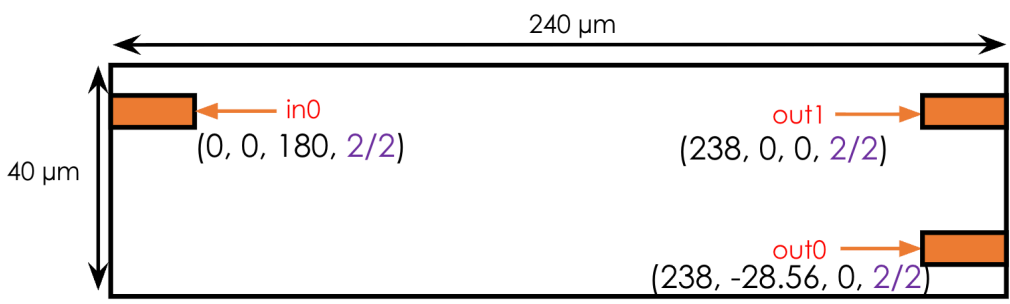
Description	
Black box name	AN800BB_Crossing_TE_C
Maturity level	Released
Required modules	X1
Dimensions	
Box size	Refer to schematic
Input and output waveguide widths	1 μm
Ports	
Optical ports	in0, in1, out0, out1
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_Crossing_TE_C	
BIB and BOT	
BIB	Refer to schematic
BOT	Refer to schematic

4.3. Crossing Balanced black box

Description	
Black box name	AN800BB_Crossing_Balanced_C
Maturity level	Alpha
Required modules	X1
Dimensions	
Box size	Refer to schematic
Input and output waveguide widths	1 μm
Ports	
Optical ports	in0, in1, out0, out1
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_Crossing_Balanced_C	
BIB and BOT	
BIB	Refer to schematic
BOT	Refer to schematic

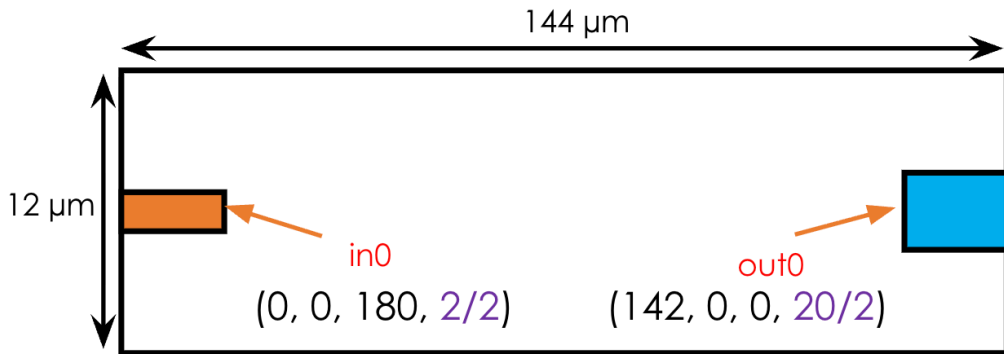
5. Polarization Management

5.1. Polarization Beam Splitter black box

Description	
Black box name	AN800BB_PBS_C
Maturity level	Beta
Required modules	X1
Dimensions	
Box size	240 μm x 40 μm
Input and output waveguide widths	1 μm
Ports	
Optical ports	in0, out0, out1
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_PBS_C	
BIB and BOT	
BIB	[-1.0, -34.5, 0]
BOT	[239.0, 5.5, 0]

6. Others

6.1. Optical Via Black Box

Description	
Black box name	AN800BB_OVIA_C
Maturity level	Beta
Required modules	X1, X2
Dimensions	
Box size	144 μm x 12 μm
Input and output waveguide widths	1 μm (in0), 2 μm (out0)
Ports	
Optical ports	in0, out0
Electrical ports	none
Schematic	
	
Modelling aspects	
The name of the BB black-box should be AN800BB_OVIA_C	
BIB and BOT	
BIB	[-1.0, -6.0, 0]
BOT	[143.0, 6.0, 0]

7. Change Log

v8.9 (03.2026), M. Aamer, L. Despax

Updated all Black Boxes dimensions

Increased ExSpot for packaging SMF fibre black box's maturity

Increased Optical Via black box's maturity

Increased Polarization Beam Splitter Via black box's maturity

Increased 1x2, 2x2 50/50, 72/28, 85/15, 90/10 black boxes maturities

Increased Crossings Compact TE and Ultra Compact black boxes maturities to released

v8.8 (09.2025), L. Despax

v8.7 (01.2025), M. Aamer

Renamed BBs

Removed AN800 v8.6 Compact Crossing black box

Removed AN800 v8.6 Polarization beam splitter MZI based black box

Removed AN800 v8.6 Polarization beam rotator black box

Added ExSpot for packaging SMF fibre black box

Removed AN800 v8.6 Directional Coupler black box

12.09.2023 (V8.6.1), M. Aamer

Restructuring of the Generic Layout Manual

05.02.2023 (V8.6), M. Aamer

New blackboxes alignment nomenclature (BIT and BOB removed)

Added new crossings BB

Added Focused GC BB

Added Optical VIA BB

Corrected pins' positions 2x2 MMI 72/28 BB

Corrected BIB/BOT for 4x4 MMI BB

18.08.2022 (V8.5.1), M.Aamer

New ports information needed (x position, y position, angle, layer/2) New layers used (module_layer/datatype=2)

New blackboxes alignment nomenclature Removed GCBB and PBS_DCBB

Added PBSBB and PRBB